2016.0 RANGE ROVER (LG), 100-00

# GENERAL INFORMATION

DESCRIPTION AND OPERATION

## ANTI-LOCK BRAKE SYSTEM CONTROL MODULE (ABS)

**CAUTION:**

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

**NOTES:**

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- DO **NOT** perform a voltage or resistance test on a wheel speed sensor when it has been removed from the vehicle. This could cause internal damage to the wheel speed sensor.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be

logged in the Anti-lock Brake System Control Module (ABS). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.
For additional information, refer to: Braking Control System (206-11 Brake Controls, Diagnosis and Testing).

| DTC | DESCRIPTION | POSSIBLE CAUSES | ACTION |
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| C0001-49 | TCS Control Channel "A" Valve 1 - Internal electronic failure | ▪ Anti-lock brake system hydraulic control unit internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| C0002-49 | TCS Control Channel "A" Valve 2 - Internal electronic failure | ▪ Anti-lock brake system hydraulic control unit internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| C0003-49 | TCS Control Channel "B" Valve 1 - Internal electronic failure | ▪ Anti-lock brake system hydraulic control unit internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| C0004-49 | TCS Control Channel "B" Valve 2 - Internal electronic failure | ▪ Anti-lock brake system hydraulic control unit internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| C0010-49 | Left Front Inlet Control - Internal electronic failure | ▪ Anti-lock brake system hydraulic control unit internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |

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| C0011-49 | Left Front Outlet Control - Internal electronic failure | ▪ Anti-lock brake system hydraulic control unit internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| C0014-49 | Right Front Inlet Control - Internal electronic failure | ▪ Anti-lock brake system hydraulic control unit internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| C0015-49 | Right Front Outlet Control - Internal electronic failure | ▪ Anti-lock brake system hydraulic control unit internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| C0018-49 | Left Rear Inlet Control - Internal electronic failure | ▪ Anti-lock brake system hydraulic control unit internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| C0019-49 | Left Rear Outlet Control - Internal electronic failure | ▪ Anti-lock brake system hydraulic control unit internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| C001C-49 | Right Rear Inlet Control - Internal electronic failure | ▪ Anti-lock brake system hydraulic control unit internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |

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| C001D-49 | Right Rear Outlet Control - Internal electronic failure | ▪ Anti-lock brake system hydraulic control unit internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| C0020-49 | ABS Pump Motor Control - Internal electronic failure | ▪ Anti-lock brake system hydraulic control unit pump power or ground circuit open circuit, high resistance<br>▪ Anti-lock brake system hydraulic control unit internal failure | ▪ Refer to the electrical circuit diagrams and check the anti-lock brake system hydraulic control unit pump power and ground circuits for open circuit, high resistance<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| C0030-38 | Left Front Tone Wheel - Signal frequency incorrect | ▪ Front left wheel speed sensor reluctor ring damaged<br>▪ Front left wheel bearing failure<br>▪ Front left wheel/tire size incorrect | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph)<br>▪ Check the integrity of the front left wheel speed sensor reluctor ring<br>▪ Check the front left wheel bearing<br>▪ Check that the wheels and tires comply with the manufacturer's specification for the vehicle. Check the front left wheel and tire for condition and balance. |

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| | | | Check and tighten the wheel nuts |
| C0033-38 | Right Front Tone Wheel - Signal frequency incorrect | ▪ Front right wheel speed sensor reluctor ring damaged<br>▪ Front right wheel bearing failure<br>▪ Front right wheel/tire size incorrect | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph)<br>▪ Check the integrity of the front right wheel speed sensor reluctor ring<br>▪ Check the front right wheel bearing<br>▪ Check that the wheels and tires comply with the manufacturer's specification for the vehicle. Check the front right wheel and tire for condition and balance. Check and tighten the wheel nuts |
| C0036-38 | Left Rear Tone Wheel - Signal frequency incorrect | ▪ Rear left wheel speed sensor reluctor ring damaged<br>▪ Rear left wheel bearing failure<br>▪ Rear left wheel/tire size incorrect | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph)<br>▪ Check the integrity of the rear left wheel speed sensor reluctor ring<br>▪ Check the rear left wheel bearing<br>▪ Check that the wheels |

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| | | | and tires comply with the manufacturer's specification for the vehicle. Check the rear left wheel and tire for condition and balance. Check and tighten the wheel nuts |
| C0039-38 | Right Rear Tone Wheel - Signal frequency incorrect | ▪ Rear right wheel speed sensor reluctor ring damaged<br>▪ Rear right wheel bearing failure<br>▪ Rear right wheel/tire size incorrect | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph)<br>▪ Check the integrity of the rear right wheel speed sensor reluctor ring<br>▪ Check the rear right wheel bearing<br>▪ Check that the wheels and tires comply with the manufacturer's specification for the vehicle. Check the rear right wheel and tire for condition and balance. Check and tighten the wheel nuts |
| C0040-64 | Brake Pedal Switch A - Signal plausibility failure | ▪ Brake pedal switch incorrectly installed<br>▪ Brake pedal switch maladjusted<br>▪ Brake pedal switch circuit short circuit to ground, short circuit to power, open circuit, high resistance | |

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| | | | **NOTE:**<br>This DTC may log in conjunction with DTC P0573-64. Check if DTC P0573-64 has been logged by the anti-lock brake system control module and, if the DTC is present, follow the prescribed remedial steps as described in the helptext for DTC P0573-64<br>▪ Check the brake pedal switch installation<br>▪ Check the brake pedal switch adjustment<br>▪ Refer to the electrical circuit diagrams and check the brake pedal switch circuit for short circuit to ground, short circuit to power, open circuit, high resistance |
| C0049-4B | Brake Fluid - Over temperature | ▪ Brake fluid over temperature | ▪ Allow the vehicle to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest |
| C0062-29 | Longitudinal Acceleration Sensor - Signal invalid | ▪ Restraints control module power or ground circuit open circuit, high resistance<br>▪ Restraints control module incorrectly installed<br>▪ Restraints system fault | ▪ Refer to the electrical circuit diagrams and check the restraints control module power and ground circuits for open circuit, high resistance<br>▪ Check the installation of the Restraints control module<br>▪ Using the manufacturer approved diagnostic |

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| | | | system, check the restraints control module for related DTCs and refer to the relevant DTC index |
| C0062-81 | Longitudinal Acceleration Sensor - Invalid serial data received | ▪ Invalid data received from another control module via the high speed CAN bus (chassis and powertrain) | ▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index |
| C0063-64 | Yaw Rate Sensor - Signal plausibility failure | ▪ Yaw rate sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Restraints system fault | **NOTES:**<br>▪ The sensor is located within/integral to the restraints control module<br>▪ This DTC can be set if the vehicle is being tested on chassis dyno rollers<br>▪ Refer to the electrical circuit diagrams and check the yaw rate sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Using the manufacturer approved diagnostic system, check the restraints control module for related DTCs and refer to the relevant DTC index |
| C0063- | Yaw Rate Sensor - | ▪ Invalid data received | ▪ Using the manufacturer |

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| 82 | Alive / sequence counter incorrect / not updated | from the restraints control module | approved diagnostic system, check the restraints control module for related DTCs and refer to the relevant DTC index |
| C0064-64 | Roll Rate Sensor - Signal plausibility failure | ▪ Roll rate sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Restraints system fault | **NOTES:**<br>▪ The sensor is located within/integral to the restraints control module<br>▪ This DTC can be set if the vehicle is being tested on chassis dyno rollers<br>▪ Refer to the electrical circuit diagrams and check the roll rate sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Using the manufacturer approved diagnostic system, check the restraints control module for related DTCs and refer to the relevant DTC index |
| C0069-81 | Yaw Rate/Longitude Sensors - Invalid serial data received | ▪ Invalid data received from the restraints control module | **NOTE:**<br>The sensor is located within/integral to the restraints control module<br>▪ Using the manufacturer approved diagnostic |

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| | | | system, check the restraints control module for related DTCs and refer to the relevant DTC index |
| C006B-00 | Stability System Active Too Long - No sub type information | ▪ Wheel speed sensor fault<br>▪ Yaw rate lateral acceleration sensor fault<br>▪ Steering angle sensor fault | ▪ Using the manufacturer approved diagnostic system, check for wheel speed sensor related DTCs and perform the relevant corrective actions<br>▪ Using the manufacturer approved diagnostic system, check for yaw rate lateral acceleration sensor related DTCs and perform the relevant corrective actions<br>▪ Using the manufacturer approved diagnostic system, check for steering angle sensor related DTCs and perform the relevant corrective actions |
| C0072-4B | Brake Temperature Too High - Over temperature | ▪ Brake disc(s) over temperature | ▪ Allow the vehicle to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest |
| C0081-00 | ABS Malfunction Indicator - No sub type information | ▪ Wheel speed sensor fault<br>▪ Yaw rate lateral acceleration sensor fault<br>▪ Steering angle sensor fault | ▪ Using the manufacturer approved diagnostic system, check for wheel speed sensor related DTCs and perform the relevant corrective actions<br>▪ Using the manufacturer approved diagnostic system, check for yaw rate lateral acceleration sensor related DTCs and perform the relevant corrective actions |

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| | | | <ul><li>Using the manufacturer approved diagnostic system, check for steering angle sensor related DTCs and perform the relevant corrective actions</li></ul> |
| C0082-48 | Brake System Malfunction Indicator - Supervision software failure | <ul><li>Anti-lock brake system control module is not correctly configured</li></ul> | <ul><li>Using the manufacturer approved diagnostic system, re-configure the anti-lock brake system control module with the latest level software. Clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module</li></ul> |
| C0500-14 | Left Front Wheel Speed Sensor - Circuit short to ground or open | <ul><li>Front left wheel speed sensor signal circuit short circuit to ground, open circuit, high resistance</li><li>Front left wheel speed sensor power circuit open circuit, high resistance</li><li>Front left wheel speed sensor internal failure</li></ul> | <ul><li>Refer to the electrical circuit diagrams and check the front left wheel speed sensor signal circuit for short circuit to ground, open circuit, high resistance. Repair circuit as required. Using the manufacturer approved diagnostic system, check datalogger signal - Left Front Wheel Speed Sensor Input (0x2B06)</li><li>Refer to the electrical circuit diagrams and check the front left wheel speed sensor power circuit for open circuit, high resistance. Repair circuit as required</li><li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left wheel speed sensor</li></ul> |
| C0501- | Left Front Wheel | <ul><li>Front left wheel</li></ul> | <ul><li>Refer to the electrical</li></ul> |

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| 66 | Speed Sensor - Signal has too many transitions / events | speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Front left wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Front left wheel/tire size incorrect | circuit diagrams and check the front left wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Refer to the electrical circuit diagrams and check the front left wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Check that the wheels and tires comply with the manufacturer's specification for the vehicle. Check the front left wheel and tire for condition and balance. Check and tighten the wheel nuts |
| C0502-11 | Left Front Wheel Speed Sensor - Circuit short to ground | ▪ Front left wheel speed sensor power circuit short circuit to ground<br>▪ Front left wheel speed sensor internal failure | ▪ Refer to the electrical circuit diagrams and check the front left wheel speed sensor power circuit for short circuit to ground. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left wheel speed sensor |
| C0503-12 | Left Front Wheel Speed Sensor - Circuit short to battery | ▪ Front left wheel speed sensor power circuit short circuit to power<br>▪ Front left wheel speed sensor internal failure | ▪ Refer to the electrical circuit diagrams and check the front left wheel speed sensor power circuit for short circuit to power. Repair circuit as required<br>▪ Using the manufacturer |

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| | | ▪ Anti-lock brake system control module internal failure | approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left wheel speed sensor<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| C0503-1C | Left Front Wheel Speed Sensor - Circuit voltage out of range | ▪ Front left wheel speed sensor power circuit short circuit to power<br>▪ Front left wheel speed sensor power circuit short circuit to another wheel speed sensor circuit<br>▪ Battery/charging system fault | ▪ Refer to the electrical circuit diagrams and check the front left wheel speed sensor power circuit for short circuit to power. Repair circuit as required<br>▪ Refer to the electrical circuit diagrams and check the front left wheel speed sensor power circuit for short circuit to another wheel speed sensor circuit. Repair circuit as required<br>▪ Check system supply voltage. Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). Refer to the relevant section of the workshop manual and test the battery and charging system |
| C0504-25 | Left Front Wheel Speed Sensor - Signal shape / waveform failure | ▪ Front left wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Front left wheel speed sensor signal | |

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| | | | circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Front left wheel speed sensor internal failure | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph)<br>▪ Refer to the electrical circuit diagrams and check the front left wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Refer to the electrical circuit diagrams and check the front left wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left wheel speed sensor |
| | C0504-29 | Left Front Wheel Speed Sensor - Signal invalid | ▪ Front left wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Front left wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) |

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| | | | resistance<br>▪ Front left wheel speed sensor internal failure | ▪ Refer to the electrical circuit diagrams and check the front left wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Refer to the electrical circuit diagrams and check the front left wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left wheel speed sensor |
| | C0504-2F | Left Front Wheel Speed Sensor - Signal erratic | ▪ Front left wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Front left wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Front left wheel speed sensor internal failure | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph)<br>▪ Refer to the electrical circuit diagrams and check the front left wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Refer to the electrical circuit diagrams and check the front left wheel speed sensor signal |

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| | | | circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left wheel speed sensor |
| C0504-31 | Left Front Wheel Speed Sensor - No signal | ▪ Front left wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Front left wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Front left wheel speed sensor internal failure | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph)<br>▪ Refer to the electrical circuit diagrams and check the front left wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Refer to the electrical circuit diagrams and check the front left wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left wheel speed sensor |

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| C0504-62 | Left Front Wheel Speed Sensor - Signal compare failure | <ul><li>Front left wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance</li><li>Front left wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance</li><li>Front left wheel speed sensor internal failure</li></ul> | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph)<ul><li>Refer to the electrical circuit diagrams and check the front left wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required</li><li>Refer to the electrical circuit diagrams and check the front left wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required</li><li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left wheel speed sensor</li></ul> |
| C0504-76 | Left Front Wheel Speed Sensor - Wrong mounting position | **NOTE:**<br>The front left wheel speed sensor distance to reluctor ring is too big<ul><li>Front left wheel</li></ul> | <ul><li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, check the integrity and positioning of the front left wheel speed sensor mounting and rectify as required</li><li>Using the manufacturer approved diagnostic</li></ul> |

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| | | speed sensor incorrectly installed<br>▪ Incorrect front left wheel speed sensor installed | system, clear the DTCs and retest. If the fault persists, install a new front left wheel speed sensor of the correct specification |
| C0505-76 | Left Front Wheel Speed Sensor Correlation - Wrong mounting position | **NOTE:**<br>The front left wheel speed sensor is showing the wheel to be rotating in the opposite direction to the others<br>▪ Front left wheel speed sensor incorrectly installed<br>▪ Incorrect front left wheel speed sensor installed | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, check the integrity and positioning of the front left wheel speed sensor mounting and rectify as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left wheel speed sensor of the correct specification |
| C0506-14 | Right Front Wheel Speed Sensor - Circuit short to ground or open | ▪ Front right wheel speed sensor signal circuit short circuit to ground, open circuit, high resistance<br>▪ Front right wheel speed sensor power circuit open circuit, high resistance<br>▪ Front right wheel speed sensor internal failure | ▪ Refer to the electrical circuit diagrams and check the front right wheel speed sensor signal circuit for short circuit to ground, open circuit, high resistance. Repair circuit as required. Using the manufacturer approved diagnostic system, check datalogger signal - Right Front Wheel Speed Sensor Input (0x2B07)<br>▪ Refer to the electrical circuit diagrams and check the front right wheel speed sensor power circuit for open circuit, high resistance. Repair circuit as required |

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| | | | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right wheel speed sensor |
| C0507-66 | Right Front Wheel Speed Sensor - Signal has too many transitions / events | ▪ Front right wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Front right wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Front right wheel/tire size incorrect | ▪ Refer to the electrical circuit diagrams and check the front right wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Refer to the electrical circuit diagrams and check the front right wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Check that the wheels and tires comply with the manufacturer's specification for the vehicle. Check the front right wheel and tire for condition and balance. Check and tighten the wheel nuts |
| C0508-11 | Right Front Wheel Speed Sensor - Circuit short to ground | ▪ Front right wheel speed sensor power circuit short circuit to ground<br>▪ Front right wheel speed sensor internal failure | ▪ Refer to the electrical circuit diagrams and check the front right wheel speed sensor power circuit for short circuit to ground. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right wheel speed sensor |

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| C0509-12 | Right Front Wheel Speed Sensor - Circuit short to battery | ▪ Front right wheel speed sensor power circuit short circuit to power<br>▪ Front right wheel speed sensor internal failure<br>▪ Anti-lock brake system control module internal failure | ▪ Refer to the electrical circuit diagrams and check the front right wheel speed sensor power circuit for short circuit to power. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right wheel speed sensor<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| C0509-1C | Right Front Wheel Speed Sensor - Circuit voltage out of range | ▪ Front right wheel speed sensor power circuit short circuit to power<br>▪ Front right wheel speed sensor power circuit short circuit to another wheel speed sensor circuit<br>▪ Battery/charging system fault | ▪ Refer to the electrical circuit diagrams and check the front right wheel speed sensor power circuit for short circuit to power. Repair circuit as required<br>▪ Refer to the electrical circuit diagrams and check the front right wheel speed sensor power circuit for short circuit to another wheel speed sensor circuit. Repair circuit as required<br>▪ Check system supply voltage. Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). Refer to the relevant section of the workshop manual and test the battery and charging system |

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| C050A-25 | Right Front Wheel Speed Sensor - Signal shape / waveform failure | <ul><li>Front right wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance</li><li>Front right wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance</li><li>Front right wheel speed sensor internal failure</li></ul> | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph)<ul><li>Refer to the electrical circuit diagrams and check the front right wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required</li><li>Refer to the electrical circuit diagrams and check the front right wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required</li><li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right wheel speed sensor</li></ul> |
| C050A-29 | Right Front Wheel Speed Sensor - Signal invalid | <ul><li>Front right wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance</li><li>Front right wheel speed sensor signal circuit short circuit to ground, short circuit to power, open</li></ul> | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph) |

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| | | circuit, high resistance<br>▪ Front right wheel speed sensor internal failure | ▪ Refer to the electrical circuit diagrams and check the front right wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Refer to the electrical circuit diagrams and check the front right wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right wheel speed sensor |
| C050A-2F | Right Front Wheel Speed Sensor - Signal erratic | ▪ Front right wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Front right wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Front right wheel speed sensor internal failure | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph)<br>▪ Refer to the electrical circuit diagrams and check the front right wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Refer to the electrical circuit diagrams and check the front right wheel speed sensor |

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| | | | | signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right wheel speed sensor | |
| | C050A-31 | Right Front Wheel Speed Sensor - No signal | ▪ Front right wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Front right wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Front right wheel speed sensor internal failure | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph)<br>▪ Refer to the electrical circuit diagrams and check the front right wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Refer to the electrical circuit diagrams and check the front right wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right wheel speed sensor | |

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| C050A-62 | Right Front Wheel Speed Sensor - Signal compare failure | ▪ Front right wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Front right wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Front right wheel speed sensor internal failure | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph)<br>▪ Refer to the electrical circuit diagrams and check the front right wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Refer to the electrical circuit diagrams and check the front right wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right wheel speed sensor |
| C050A-76 | Right Front Wheel Speed Sensor - Wrong mounting position | **NOTE:**<br>The front right wheel speed sensor distance to reluctor ring is too big<br>▪ Front right wheel | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, check the integrity and positioning of the front right wheel speed sensor mounting and rectify as required<br>▪ Using the manufacturer approved diagnostic |

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| | | <ul><li>speed sensor incorrectly installed</li><li>Incorrect front right wheel speed sensor installed</li></ul> | system, clear the DTCs and retest. If the fault persists, install a new front right wheel speed sensor of the correct specification |
| C050B-76 | Right Front Wheel Speed Sensor Correlation - Wrong mounting position | **NOTE:**<br>The front right wheel speed sensor is showing the wheel to be rotating in the opposite direction to the others<br><ul><li>Front right wheel speed sensor incorrectly installed</li><li>Incorrect front right wheel speed sensor installed</li></ul> | <ul><li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, check the integrity and positioning of the front right wheel speed sensor mounting and rectify as required</li><li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right wheel speed sensor of the correct specification</li></ul> |
| C050C-14 | Left Rear Wheel Speed Sensor - Circuit short to ground or open | <ul><li>Rear left wheel speed sensor signal circuit short circuit to ground, open circuit, high resistance</li><li>Rear left wheel speed sensor power circuit open circuit, high resistance</li><li>Rear left wheel speed sensor internal failure</li></ul> | <ul><li>Refer to the electrical circuit diagrams and check the rear left wheel speed sensor signal circuit for short circuit to ground, open circuit, high resistance. Repair circuit as required. Using the manufacturer approved diagnostic system, check datalogger signal - Left Rear Wheel Speed Sensor Input (0x2B08)</li><li>Refer to the electrical circuit diagrams and check the rear left wheel speed sensor power circuit for open circuit, high resistance. Repair circuit as required</li></ul> |

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| | | | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear left wheel speed sensor |
| C050D-66 | Left Rear Wheel Speed Sensor - Signal has too many transitions / events | ▪ Rear left wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Rear left wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Rear left wheel/tire size incorrect | ▪ Refer to the electrical circuit diagrams and check the rear left wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Refer to the electrical circuit diagrams and check the rear left wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Check that the wheels and tires comply with the manufacturer's specification for the vehicle. Check the rear left wheel and tire for condition and balance. Check and tighten the wheel nuts |
| C050E-11 | Left Rear Wheel Speed Sensor - Circuit short to ground | ▪ Rear left wheel speed sensor power circuit short circuit to ground<br>▪ Rear left wheel speed sensor internal failure | ▪ Refer to the electrical circuit diagrams and check the rear left wheel speed sensor power circuit for short circuit to ground. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear left wheel speed sensor |

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| C050F-12 | Left Rear Wheel Speed Sensor - Circuit short to battery | ▪ Rear left wheel speed sensor power circuit short circuit to power<br>▪ Rear left wheel speed sensor internal failure<br>▪ Anti-lock brake system control module internal failure | ▪ Refer to the electrical circuit diagrams and check the rear left wheel speed sensor power circuit for short circuit to power. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear left wheel speed sensor<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| C050F-1C | Left Rear Wheel Speed Sensor - Circuit voltage out of range | ▪ Rear left wheel speed sensor power circuit short circuit to power<br>▪ Rear left wheel speed sensor power circuit short circuit to another wheel speed sensor circuit<br>▪ Battery/charging system fault | ▪ Refer to the electrical circuit diagrams and check the rear left wheel speed sensor power circuit for short circuit to power. Repair circuit as required<br>▪ Refer to the electrical circuit diagrams and check the rear left wheel speed sensor power circuit for short circuit to another wheel speed sensor circuit. Repair circuit as required<br>▪ Check system supply voltage. Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). Refer to the relevant section of the workshop manual and test the battery and charging system |
| C0510-25 | Left Rear Wheel Speed Sensor - Signal | ▪ Rear left wheel speed sensor power | |

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| | shape / waveform failure | circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Rear left wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Rear left wheel speed sensor internal failure | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph)<br>▪ Refer to the electrical circuit diagrams and check the rear left wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Refer to the electrical circuit diagrams and check the rear left wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear left wheel speed sensor |
| C0510-29 | Left Rear Wheel Speed Sensor - Signal invalid | ▪ Rear left wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Rear left wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph)<br>▪ Refer to the electrical |

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| | | | <ul><li>Rear left wheel speed sensor internal failure</li></ul> | circuit diagrams and check the rear left wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<ul><li>Refer to the electrical circuit diagrams and check the rear left wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required</li><li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear left wheel speed sensor</li></ul> |
| | C0510-2F | Left Rear Wheel Speed Sensor - Signal erratic | <ul><li>Rear left wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance</li><li>Rear left wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance</li><li>Rear left wheel speed sensor internal failure</li></ul> | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph)<ul><li>Refer to the electrical circuit diagrams and check the rear left wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required</li><li>Refer to the electrical circuit diagrams and check the rear left wheel speed sensor signal circuit for short circuit to ground, short circuit to</li></ul> |

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| | | | power, open circuit, high resistance. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear left wheel speed sensor |
| C0510-31 | Left Rear Wheel Speed Sensor - No signal | ▪ Rear left wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Rear left wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Rear left wheel speed sensor internal failure | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph)<br>▪ Refer to the electrical circuit diagrams and check the rear left wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Refer to the electrical circuit diagrams and check the rear left wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear left wheel speed sensor |
| C0510-62 | Left Rear Wheel Speed Sensor - Signal | ▪ Rear left wheel speed sensor power | |

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| | | compare failure | circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Rear left wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Rear left wheel speed sensor internal failure | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph)<br>▪ Refer to the electrical circuit diagrams and check the rear left wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Refer to the electrical circuit diagrams and check the rear left wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear left wheel speed sensor |
| | C0510-76 | Left Rear Wheel Speed Sensor - Wrong mounting position | **NOTE:**<br>The rear left wheel speed sensor distance to reluctor ring is too big<br>▪ Rear left wheel speed sensor incorrectly installed | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, check the integrity and positioning of the rear left wheel speed sensor mounting and rectify as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear |

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| | | ▪ Incorrect rear left wheel speed sensor installed | left wheel speed sensor of the correct specification |
| C0511-76 | Left Rear Wheel Speed Sensor Correlation - Wrong mounting position | **NOTE:** The rear left wheel speed sensor is showing the wheel to be rotating in the opposite direction to the others<br>▪ Rear left wheel speed sensor incorrectly installed<br>▪ Incorrect rear left wheel speed sensor installed | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, check the integrity and positioning of the rear left wheel speed sensor mounting and rectify as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear left wheel speed sensor of the correct specification |
| C0512-14 | Right Rear Wheel Speed Sensor - Circuit short to ground or open | ▪ Rear right wheel speed sensor signal circuit short circuit to ground, open circuit, high resistance<br>▪ Rear right wheel speed sensor power circuit open circuit, high resistance<br>▪ Rear right wheel speed sensor internal failure | ▪ Refer to the electrical circuit diagrams and check the rear right wheel speed sensor signal circuit for short circuit to ground, open circuit, high resistance. Repair circuit as required. Using the manufacturer approved diagnostic system, check datalogger signal - Right Rear Wheel Speed Sensor Input (0x2B09)<br>▪ Refer to the electrical circuit diagrams and check the rear right wheel speed sensor power circuit for open circuit, high resistance. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs |

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| | | | and retest. If the fault persists, install a new rear right wheel speed sensor |
| C0513-66 | Right Rear Wheel Speed Sensor - Signal has too many transitions / events | ▪ Rear right wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Rear right wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Rear right wheel/tire size incorrect | ▪ Refer to the electrical circuit diagrams and check the rear right wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Refer to the electrical circuit diagrams and check the rear right wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Check that the wheels and tires comply with the manufacturer's specification for the vehicle. Check the rear right wheel and tire for condition and balance. Check and tighten the wheel nuts |
| C0514-11 | Right Rear Wheel Speed Sensor - Circuit short to ground | ▪ Rear right wheel speed sensor power circuit short circuit to ground<br>▪ Rear right wheel speed sensor internal failure | ▪ Refer to the electrical circuit diagrams and check the rear right wheel speed sensor power circuit for short circuit to ground. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear right wheel speed sensor |
| C0515-12 | Right Rear Wheel Speed Sensor - Circuit short to | ▪ Rear right wheel speed sensor power circuit short circuit to | ▪ Refer to the electrical circuit diagrams and check the rear right |

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| | battery | power<br>▪ Rear right wheel speed sensor internal failure<br>▪ Anti-lock brake system control module internal failure | wheel speed sensor power circuit for short circuit to power. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear right wheel speed sensor<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| C0515-1C | Right Rear Wheel Speed Sensor - Circuit voltage out of range | ▪ Rear right wheel speed sensor power circuit short circuit to power<br>▪ Rear right wheel speed sensor power circuit short circuit to another wheel speed sensor circuit<br>▪ Battery/charging system fault | ▪ Refer to the electrical circuit diagrams and check the rear right wheel speed sensor power circuit for short circuit to power. Repair circuit as required<br>▪ Refer to the electrical circuit diagrams and check the rear right wheel speed sensor power circuit for short circuit to another wheel speed sensor circuit. Repair circuit as required<br>▪ Check system supply voltage. Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). Refer to the relevant section of the workshop manual and test the battery and charging system |
| C0516-25 | Right Rear Wheel Speed Sensor - Signal shape / waveform failure | ▪ Rear right wheel speed sensor power circuit short circuit to ground, short circuit to power, open | |

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| | | | circuit, high resistance<br>▪ Rear right wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Rear right wheel speed sensor internal failure | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph)<br>▪ Refer to the electrical circuit diagrams and check the rear right wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Refer to the electrical circuit diagrams and check the rear right wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear right wheel speed sensor |
| | C0516-29 | Right Rear Wheel Speed Sensor - Signal invalid | ▪ Rear right wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Rear right wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph)<br>▪ Refer to the electrical |

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| | | ▪ Rear right wheel speed sensor internal failure | circuit diagrams and check the rear right wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Refer to the electrical circuit diagrams and check the rear right wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear right wheel speed sensor |
| C0516-2F | Right Rear Wheel Speed Sensor - Signal erratic | ▪ Rear right wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Rear right wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Rear right wheel speed sensor internal failure | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph)<br>▪ Refer to the electrical circuit diagrams and check the rear right wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Refer to the electrical circuit diagrams and check the rear right wheel speed sensor signal circuit for short circuit to ground, short |

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| | | | circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear right wheel speed sensor |
| C0516-31 | Right Rear Wheel Speed Sensor - No signal | ▪ Rear right wheel speed sensor power circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Rear right wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Rear right wheel speed sensor internal failure | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph)<br>▪ Refer to the electrical circuit diagrams and check the rear right wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Refer to the electrical circuit diagrams and check the rear right wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear right wheel speed sensor |
| C0516-62 | Right Rear Wheel Speed Sensor - Signal | ▪ Rear right wheel speed sensor power | |

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| | compare failure | circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Rear right wheel speed sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Rear right wheel speed sensor internal failure | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph)<br>▪ Refer to the electrical circuit diagrams and check the rear right wheel speed sensor power circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Refer to the electrical circuit diagrams and check the rear right wheel speed sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear right wheel speed sensor |
| C0516-76 | Right Rear Wheel Speed Sensor - Wrong mounting position | **NOTE:**<br>The rear right wheel speed sensor distance to reluctor ring is too big<br>▪ Rear right wheel speed sensor incorrectly installed | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, check the integrity and positioning of the rear right wheel speed sensor mounting and rectify as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear |

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| | | ▪ Incorrect rear right wheel speed sensor installed | right wheel speed sensor of the correct specification |
| C0517-76 | Right Rear Wheel Speed Sensor Correlation - Wrong mounting position | **NOTE:**<br>The rear right wheel speed sensor is showing the wheel to be rotating in the opposite direction to the others<br>▪ Rear right wheel speed sensor incorrectly installed<br>▪ Incorrect rear right wheel speed sensor installed | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, check the integrity and positioning of the rear right wheel speed sensor mounting and rectify as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear right wheel speed sensor of the correct specification |
| C051C-64 | Lateral Acceleration Sensor - Signal plausibility failure | ▪ Lateral acceleration sensor internal failure | ▪ Using the manufacturer approved diagnostic system, check the restraints control module for related DTCs and refer to the relevant DTC index |
| C0526-81 | Steering Angle Sensor Module - Invalid serial data received | ▪ Invalid data received from the steering angle sensor module | ▪ Using the manufacturer approved diagnostic system, check the steering angle sensor module for related DTCs and refer to the relevant DTC index |
| C0528-64 | Steering Angle Sensor - Signal plausibility failure | ▪ Yaw rate sensor incorrect installation<br>▪ Steering angle sensor incorrectly installed<br>▪ Steering angle | |

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| | | sensor internal failure | **NOTE:**<br>This DTC is set when there is conflicting information coming from the steering angle sensor and the yaw rate sensor<br>▪ Check the yaw rate sensor installation and rectify as required<br>▪ Using the manufacturer approved diagnostic system, check for steering angle sensor related DTCs and refer to the relevant DTC index<br>▪ Check the operation of the steering angle sensor. If faulty, install a new steering angle sensor module as required | |
| | C052A-64 | Steering Angle Sensor Module Correlation - Signal plausibility failure | ▪ Yaw rate sensor incorrect installation<br>▪ Steering angle sensor incorrectly installed<br>▪ Steering angle sensor internal failure | **NOTE:**<br>This DTC is set when there is conflicting information coming from the steering angle sensor and the yaw rate sensor<br>▪ Check the yaw rate sensor installation and rectify as required<br>▪ Using the manufacturer approved diagnostic system, check for steering angle sensor related DTCs and refer to the relevant DTC index<br>▪ Check the operation of |

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| | | | the steering angle sensor. If faulty, install a new steering angle sensor module as required |
| C052A-92 | Steering Angle Sensor - Performance or incorrect operation | ▪ Steering angle sensor incorrectly installed<br>▪ Steering angle sensor internal failure | ▪ Using the manufacturer approved diagnostic system, check for steering angle sensor related DTCs and refer to the relevant DTC index<br>▪ Check the operation of the steering angle sensor. If faulty, install a new steering angle sensor module as required |
| C052B-49 | ABS Pump Motor Control Range/Performance - Internal electronic failure | ▪ Anti-lock brake system hydraulic control unit pump power or ground circuit open circuit, high resistance<br>▪ Anti-lock brake system control module internal failure | ▪ Refer to the electrical circuit diagrams and check the anti-lock brake system hydraulic control unit pump power and ground circuits for open circuit, high resistance<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| C052C-16 | ABS Pump Motor Control Circuit/Open - Circuit voltage below threshold | ▪ Anti-lock brake system hydraulic control unit pump circuit short circuit to ground, short circuit to power, open circuit, high resistance | ▪ Refer to the electrical circuit diagrams and check the anti-lock brake system hydraulic control unit pump circuit for short circuit to ground, short circuit to power, open circuit, high resistance |
| C052E-16 | ABS Pump Motor Control Circuit Low - Circuit voltage below threshold | ▪ Anti-lock brake system hydraulic control unit pump power or ground circuit open circuit, high resistance | ▪ Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). Refer to the electrical circuit diagrams and |

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| | | ▪ Battery/charging system fault | check the anti-lock brake system hydraulic control unit pump power and ground circuits for open circuit, high resistance<br>▪ Refer to the relevant section of the workshop manual and test the battery and charging system |
| C0533-14 | Return Pump - Circuit short to ground or open | ▪ Anti-lock brake system hydraulic control unit pump circuit(s) short circuit to ground, open circuit, high resistance<br>▪ Anti-lock brake system control module internal failure | ▪ Refer to the electrical circuit diagrams and check the anti-lock brake system hydraulic control unit pump supply and the anti-lock brake system control module ground circuits for short circuit to ground, open circuit, high resistance. Repair circuit as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| C0533-16 | Return Pump - Circuit voltage below threshold | ▪ Anti-lock brake system hydraulic control unit pump power or ground circuit open circuit, high resistance<br>▪ Battery/charging system fault | ▪ Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). Refer to the electrical circuit diagrams and check the anti-lock brake system hydraulic control unit pump power and ground circuits for open circuit, high resistance<br>▪ Refer to the relevant section of the workshop manual and test the battery and charging system |
| C0533- | Motor Control "A" | ▪ Anti-lock brake | ▪ Refer to the electrical |

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| 49 | Circuit Low - Internal electronic failure | system hydraulic control unit pump power or ground circuit open circuit, high resistance<br>▪ Anti-lock brake system control module internal failure | circuit diagrams and check the anti-lock brake system hydraulic control unit pump power and ground circuits for open circuit, high resistance<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| C053B-16 | Control Module Input Power "A" - Circuit voltage below threshold | ▪ Anti-lock brake system hydraulic control unit pump power or ground circuit open circuit, high resistance<br>▪ Battery/charging system fault | ▪ Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). Refer to the electrical circuit diagrams and check the anti-lock brake system hydraulic control unit pump power and ground circuits for open circuit, high resistance<br>▪ Refer to the relevant section of the workshop manual and test the battery and charging system |
| C053C-49 | Wheel Speed Sensor Generic Range/Performance - Internal electronic failure | ▪ Battery/charging system fault<br>▪ Wheel speed sensor signal circuit(s) short circuit to ground, short circuit to power, short circuit to another sensor circuit, open circuit, high resistance<br>▪ Wheel speed sensor(s) internal failure | ▪ Refer to the relevant section of the workshop manual and test the battery and charging system<br>▪ Refer to the electrical circuit diagrams and check the wheel speed sensor signal circuits for short circuit to ground, short circuit to power, short circuit to another sensor circuit, open circuit, high resistance. Repair circuit(s) as required<br>▪ Using the manufacturer approved diagnostic |

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| | | | system, clear the DTCs and retest. If the fault persists, install new wheel speed sensor(s) |
| C053C-76 | Wheel Speed Sensor - Wrong mounting position | ▪ Wheel speed sensor(s) incorrectly installed<br>▪ Incorrect wheel speed sensor(s) installed | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, check the integrity and positioning of the wheel speed sensor mounting(s) and rectify as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install new wheel speed sensor(s) of the correct specification |
| C053D-28 | Pressure Sensor - Signal bias level Out of range / zero adjustment failure | ▪ Anti-lock brake system hydraulic control unit internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| C053E-14 | Pressure Sensor - Circuit short to ground or open | ▪ Anti-lock brake system hydraulic control unit internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| C053F-14 | Brake Pressure Sensor "A" Circuit High - Circuit short to ground or open | ▪ Anti-lock brake system hydraulic control unit internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| C0552-64 | Longitudinal Acceleration Sensor - | ▪ Longitudinal acceleration sensor | |

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| | Signal plausibility failure | internal failure | **NOTE:**<br>This DTC can be set if the vehicle is being tested on chassis dyno rollers<br>▪ Using the manufacturer approved diagnostic system, check the restraints control module for related DTCs and refer to the relevant DTC index |
| C0555-4A | Left Front Wheel Speed Sensor - Incorrect component installed | ▪ Incorrect front left wheel speed sensor installed | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front left wheel speed sensor of the correct specification |
| C0556-4A | Right Front Wheel Speed Sensor - Incorrect component installed | ▪ Incorrect front right wheel speed sensor installed | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front right wheel speed sensor of the correct specification |
| C0557-4A | Left Rear Wheel Speed Sensor - Incorrect component installed | ▪ Incorrect rear left wheel speed sensor installed | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear left wheel speed sensor of the correct specification |
| C0558-4A | Right Rear Wheel Speed Sensor - Incorrect component installed | ▪ Incorrect rear right wheel speed sensor installed | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear |

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| | | | right wheel speed sensor of the correct specification |
| C055B-16 | ABS Pump Motor Supply Voltage Low - Circuit voltage below threshold | ▪ Anti-lock brake system hydraulic control unit pump power or ground circuit open circuit, high resistance<br>▪ Battery/charging system fault | ▪ Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). Refer to the electrical circuit diagrams and check the anti-lock brake system hydraulic control unit pump power and ground circuits for open circuit, high resistance<br>▪ Refer to the relevant section of the workshop manual and test the battery and charging system |
| C0563-00 | ABS Control Module Performance - No sub type information | ▪ Wheel bearing failure<br>▪ Wheel speed sensor reluctor ring failure<br>▪ High speed CAN bus (chassis or powertrain) circuits short circuit to ground, short circuit to power, open circuit, high resistance | **NOTE:**<br>If this DTC is logged for the first time, the only action is to clear the DTC. Should the DTC reoccur, follow the actions as described below<br>▪ Check the integrity of the wheel bearings<br>▪ Using the manufacturer approved diagnostic system, check datalogger signals - Left Front Wheel Speed Sensor Input (0x2B06) - Right Front Wheel Speed Sensor Input (0x2B07) - Left Rear Wheel Speed Sensor Input (0x2B08) - Right Rear Wheel Speed Sensor Input (0x2B09) |

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| | | | | ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (chassis and powertrain) circuits for short circuit to ground, short circuit to power, open circuit, high resistance | |
| | C0564-16 | ABS Control Module System Voltage Low - Circuit voltage below threshold | ▪ Anti-lock brake system control module power or ground circuit open circuit, high resistance<br>▪ Battery/charging system fault | ▪ Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). Refer to the electrical circuit diagrams and check the anti-lock brake system control module power and ground circuits for open circuit, high resistance<br>▪ Refer to the relevant section of the workshop manual and test the battery and charging system | |
| | C0565-17 | Battery Voltage - Circuit voltage above threshold | ▪ Battery/charging system fault | ▪ Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). Refer to the relevant section of the workshop manual and test the battery and charging system | |
| | C1054-68 | System Functionality Reduced By Diagnostic Request - Event information | ▪ Diagnostic routine DC00 in progress | | |

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| | | | **NOTE:**<br><br>This DTC is set when routine DC00 is performed (Engineering only)<br><br>- Exit the diagnostic routine |
| C105C-68 | ABS Dynomometer Test Mode Active - Event information | - Dynomometer test mode active | |

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| | | | NOTES:<br><br>- Dynomometer test mode is entered when the accelerator pedal, brake pedal and dynamic stability control switch are simultaneously pressed for more than 5 seconds when the vehicle is not moving and the steering is in the straight ahead position. The anti-lock brake system and dynamic stability control warning indicators are also illuminated<br>- Dynomometer test mode is exited by briefly pressing the dynamic stability control switch or turning the steering wheel through more than 90° (when the ignition is set to on). The anti-lock brake system and dynamic stability control warning indicators will extinguish<br><br>- Exit dynomometer test mode | |
| C1061-06 | Brake Disc Wear - Algorithm based failures | - Anti-lock brake system control module internal failure | - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault | |

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| | | | persists, install a new anti-lock brake system control module |
| C1067-24 | All Terrain / Road Progress Control Switch - Signal stuck high | ▪ All terrain progress control switch stuck active | ▪ Test the operation of the all terrain progress control switch. If fault persists, install a new terrain response switchpack |
| C1109-24 | Vehicle Dynamics Control Switch - Signal stuck high | ▪ Dynamic stability control switch stuck active | ▪ Test the operation of the dynamic stability control switch. If fault persists, install a new terrain response switchpack |
| C1A8A-95 | Front Axle Wheel Speed Sensors Swapped - Incorrect assembly | ▪ Front wheel speed sensor circuits transposed | ▪ Refer to the electrical circuit diagrams and check the front wheel speed sensor circuits for correct connection to the anti-lock brake system control module |
| C1A8B-95 | Rear Axle Wheel Speed Sensors Swapped - Incorrect assembly | ▪ Rear wheel speed sensor circuits transposed | ▪ Refer to the electrical circuit diagrams and check the rear wheel speed sensor circuits for correct connection to the anti-lock brake system control module |
| C1A8C-64 | Wheel Speed Sensor Supply Circuit - Signal plausibility failure | ▪ Wheel speed sensor signal circuit(s) short circuit to ground, short circuit to power, short circuit to another sensor circuit, open circuit, high resistance<br>▪ Wheel speed sensor reluctor ring fault<br>▪ Wheel bearing failure<br>▪ Wheel and/or tire size incorrect | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph)<br>▪ Refer to the electrical circuit diagrams and check the wheel speed |

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| | | | | sensor signal circuits for short circuit to ground, short circuit to power, short circuit to another sensor circuit, open circuit, high resistance. Repair circuit(s) as required<br>▪ Using the manufacturer approved diagnostic system, check datalogger signals - Left Front Wheel Speed Sensor Input (0x2B06) - Right Front Wheel Speed Sensor Input (0x2B07) - Left Rear Wheel Speed Sensor Input (0x2B08) - Right Rear Wheel Speed Sensor Input (0x2B09)<br>▪ Check the integrity of the wheel bearings<br>▪ Check that the wheel and tire sizes are correct to the vehicle specification | |
| | C1A8D-64 | Wheel Speed Sensor - Signal plausibility failure | ▪ Wheel speed sensor signal circuit(s) short circuit to ground, short circuit to power, short circuit to another sensor circuit, open circuit, high resistance<br>▪ Wheel speed sensor reluctor ring fault<br>▪ Wheel bearing failure<br>▪ Wheel and/or tire size incorrect | **NOTE:**<br>After clearing the DTCs, the warning indicator(s) may not extinguish until the vehicle speed has exceeded 10mph (15kph)<br>▪ Refer to the electrical circuit diagrams and check the wheel speed sensor signal circuits for short circuit to ground, short circuit to power, short circuit to another sensor circuit, open circuit, high resistance. Repair circuit(s) as required<br>▪ Using the manufacturer | |

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| | | | approved diagnostic system, check datalogger signals - Left Front Wheel Speed Sensor Input (0x2B06) - Right Front Wheel Speed Sensor Input (0x2B07) - Left Rear Wheel Speed Sensor Input (0x2B08) - Right Rear Wheel Speed Sensor Input (0x2B09)<br>▪ Check the integrity of the wheel bearings<br>▪ Check that the wheel and tire sizes are correct to the vehicle specification |
| C1A8E-12 | Wheel Speed Sensor Supply - Circuit short to battery | ▪ Wheel speed sensor supply circuit(s) short circuit to power<br>▪ Anti-lock brake system control module internal failure | ▪ Refer to the electrical circuit diagrams and check the wheel speed sensor supply circuits for short circuit to power. Repair circuit(s) as required<br>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| C1B22-24 | Hill Descent Switch - Signal stuck high | ▪ Hill descent control switch stuck active | ▪ Test the operation of the hill descent control switch. If fault persists, install a new terrain response switchpack |
| P0563-17 | System Voltage High - Circuit voltage above threshold | ▪ Battery/charging system fault | ▪ Refer to the relevant section of the workshop manual and test the battery and charging system |
| P0573-64 | Brake Switch A Circuit High - Signal plausibility failure | ▪ Brake pedal switch incorrectly installed<br>▪ Brake pedal switch | |

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| | | | maladjusted<br>- Brake pedal switch circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>- Irregular use of brake pedal during driving | **NOTE:**<br>This DTC will log if the brake pedal switch is continuously active for 45 seconds or more while the vehicle speed is 17kph or greater and throttle application is at 5% or higher. If all these conditions continue, the Dynamic Stability Control (DSC) warning indicator will illuminate, under some conditions further warning indicators may be activated and a warning message may appear in the message centre. This DTC may log in conjunction with DTC C0040-64<br>- Check the brake pedal switch installation<br>- Check the brake pedal switch adjustment<br>- Refer to the electrical circuit diagrams and check the brake pedal switch circuit for short circuit to ground, short circuit to power, open circuit, high resistance<br>- If there are no faults with the brake pedal switch or circuits, clear the DTC and roadtest the vehicle. To fully test for a brake pedal switch fault during the roadtest, it is necessary to ensure that the vehicle is driven over | |

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| | | | 17kph without the brakes being applied for longer than one minute. If driving conditions demand use of the brake during this 60 second period, the brakes may be applied as required for safe driving, however the brake pedal switch test will need to be restarted<br>- If the fault does not occur again during the roadtest, this may indicate that the DTC is being logged as a result of the customer's driving behaviour, which may involve either left-foot braking or a foot being rested on the brake pedal during driving. Advise the customer that neither of these practices are recommended and that they should adjust their driving style accordingly |
| P05E0-81 | Brake Pedal Position Sensor "A"/"B" Correlation - Invalid serial data received | - Invalid data received from another control module via the high speed CAN bus (chassis and powertrain) | - Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index |
| P05E0-87 | Brake Pedal Position Sensor "A"/"B" Correlation - Missing message | - Missing message from another control module via the high speed CAN bus (chassis or powertrain) | - Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and |

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| | | | refer to the relevant DTC index |
| P0601-49 | Control Module - Internal electronic failure | ▪ Anti-lock brake system control module internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| P0602-49 | Control Module Programming Error - Internal electronic failure | ▪ Anti-lock brake system control module internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| P0604-49 | Internal Control Module Random Access Memory (RAM) Error - Internal electronic failure | ▪ Anti-lock brake system control module internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| P0605-49 | Internal Control Module Read Only Memory (ROM) Error - Internal electronic failure | ▪ Anti-lock brake system control module internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| P0606-49 | Control Module Processor - Internal electronic failure | ▪ Anti-lock brake system control module internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| P0607-49 | Control Module Performance - Internal electronic failure | ▪ Anti-lock brake system control module internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new |

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| | | | anti-lock brake system control module |
| P060A-48 | Internal Control Module Monitoring Processor Performance - Supervision software failure | ▪ Anti-lock brake system control module is not configured correctly | ▪ Using the manufacturer approved diagnostic system, reconfigure the anti-lock brake system control module with the latest level software. Clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| P060B-49 | Internal Control Module A/D Processing Performance - Internal electronic failure | ▪ Anti-lock brake system control module internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| P060C-49 | Internal Control Module Main Processor Performance - Internal electronic failure | ▪ Anti-lock brake system control module internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| P062F-48 | Control Module - Supervision software failure | ▪ Anti-lock brake system control module is not configured correctly | ▪ Using the manufacturer approved diagnostic system, reconfigure the anti-lock brake system control module with the latest level software. Clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| P0634-4B | Control Module Internal Temperature "A" Too High - Over temperature | ▪ Anti-lock brake system hydraulic control unit internal temperature above threshold | ▪ Allow the anti-lock brake system hydraulic control unit to cool, clear the DTC and retest. Do not renew the anti-lock brake system control module as |

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| | | ▪ Diagnostic tool is activating the anti-lock brake system hydraulic control unit valves for prolonged periods of time | this is a protection function to ensure no internal damage occurs |
| P064F-55 | Control Module Main Calibration Data - Not configured | ▪ Anti-lock brake system control module is not configured correctly | ▪ Using the manufacturer approved diagnostic system, re-configure the anti-lock brake system control module with the latest level software |
| P16A1-49 | ABS Valve Drive Fault - Internal electronic failure | ▪ Anti-lock brake system hydraulic control unit internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |
| P16A2-16 | ABS Internal Control Module Monitor Performance - Circuit voltage below threshold | ▪ Anti-lock brake system control module power or ground circuit open circuit, high resistance<br>▪ Battery/charging system fault | ▪ Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). Refer to the electrical circuit diagrams and check the anti-lock brake system control module power and ground circuits for open circuit, high resistance<br>▪ Refer to the relevant section of the workshop manual and test the battery and charging system |
| P16A2-49 | ABS Internal Control Module Monitor Performance - Internal electronic failure | ▪ Anti-lock brake system control module internal failure | ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |

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| U0001-81 | High Speed CAN Communication Bus - Invalid serial data received | ▪ Invalid data received from another control module via the high speed CAN bus (chassis and powertrain) | **NOTE:** Do NOT install a new ABS module as this will NOT rectify the fault<br>▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index |
| U0001-82 | High Speed CAN Communication Bus - Alive/sequence counter incorrect / not updated | ▪ Invalid data received from another control module via the high speed CAN bus (chassis and powertrain) | **NOTE:** Do NOT install a new ABS module as this will NOT rectify the fault<br>▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index |
| U0001-83 | High Speed CAN Communication Bus - Value of signal protection calculation incorrect | ▪ Invalid data received from another control module via the high speed CAN bus (chassis and powertrain) | **NOTE:** Do NOT install a new ABS module as this will NOT rectify the fault |

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| | | | ▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index |
| U0001-87 | High Speed CAN Communication Bus - Missing message | ▪ Missing message from another control module via the high speed CAN bus (chassis and powertrain) | **NOTE:** Do NOT install a new ABS module as this will NOT rectify the fault<br>▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index |
| U0002-87 | High Speed CAN Communication Bus Performance - Missing message | ▪ Missing message from another control module via the high speed CAN bus (chassis or powertrain) | ▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index |
| U0073-88 | High Speed CAN Communication Bus - Bus off | ▪ High speed CAN bus (chassis or powertrain) circuits short circuit to ground, short circuit | ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical |

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| | | to power, open circuit, high resistance | circuit diagrams and check the high speed CAN bus (chassis and powertrain) circuits for short circuit to ground, short circuit to power, open circuit, high resistance |
| U0074-88 | Control Module Communication Bus "B" Off - Bus off | ▪ Anti-lock brake system control module failure<br>▪ High speed CAN bus (chassis or powertrain) circuits short circuit to ground, short circuit to power, open circuit, high resistance | ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (chassis and powertrain) circuits for short circuit to ground, short circuit to power, open circuit, high resistance |
| U0100-87 | Lost Communication With ECM/PCM "A" - Missing message | ▪ Missing message from the engine control module | ▪ Using the manufacturer approved diagnostic system, check the snapshot data for details of the missing message. Check the engine control module for related DTCs and refer to the relevant DTC index |
| U0101-87 | Lost Communication with TCM - Missing message | ▪ Missing message from the transmission control module | ▪ Using the manufacturer approved diagnostic system, check the snapshot data for details of the missing message. Check the transmission control module for related DTCs and refer to the relevant DTC index |
| U0102-87 | Lost Communication with Transfer Case Control Module - Missing message | ▪ Missing message from the transfer case control module | ▪ Using the manufacturer approved diagnostic system, check the snapshot data for details of the missing message. Check the transfer case control module for |

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| | | | related DTCs and refer to the relevant DTC index |
| U0103-87 | Lost Communication With Gear Shift Control Module A - Missing message | ▪ Missing message from the transmission control switch | ▪ Using the manufacturer approved diagnostic system, check the snapshot data for details of the missing message. Check the transmission control switch for related DTCs and refer to the relevant DTC index |
| U0114-87 | Lost Communication With Four-Wheel Drive Clutch Control Module - Missing message | ▪ Missing message from the transfer case control module | ▪ Using the manufacturer approved diagnostic system, check the snapshot data for details of the missing message. Check the transfer case control module for related DTCs and refer to the relevant DTC index |
| U0115-87 | Lost Communication With ECM/PCM "B" - Missing message | ▪ Missing message from the engine control module | ▪ Using the manufacturer approved diagnostic system, check the snapshot data for details of the missing message. Check the engine control module for related DTCs and refer to the relevant DTC index |
| U0123-87 | Lost Communication With Yaw Rate Sensor Module - Missing message | ▪ Missing message from the restraints control module | ▪ Using the manufacturer approved diagnostic system, check the snapshot data for details of the missing message. Check the restraints control module for related DTCs and refer to the relevant DTC index |
| U0125-87 | Lost Communication With Multi-axis Acceleration Sensor Module - Missing message | ▪ Missing message from the restraints control module | ▪ Using the manufacturer approved diagnostic system, check the snapshot data for details of the missing message. Check the restraints |

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| | | | control module for related DTCs and refer to the relevant DTC index |
| U0126-87 | Lost Communication With Steering Angle Sensor Module - Missing message | ▪ Missing message from the steering angle sensor module | ▪ Using the manufacturer approved diagnostic system, check the snapshot data for details of the missing message. Check the steering angle sensor module for related DTCs and refer to the relevant DTC index |
| U012A-87 | Lost Communication with Chassis Control Module "A" - Missing message | ▪ Missing message from the integrated suspension control module | ▪ Using the manufacturer approved diagnostic system, check the snapshot data for details of the missing message. Check the integrated suspension control module for related DTCs and refer to the relevant DTC index |
| U0136-87 | Lost Communication With Differential Control Module - Rear - Missing message | ▪ Missing message from the rear differential control module | ▪ Using the manufacturer approved diagnostic system, check the snapshot data for details of the missing message. Check the rear differential control module for related DTCs and refer to the relevant DTC index |
| U0140-87 | Lost Communication With Body Control Module - Missing message | ▪ Missing message from the body control module/gateway module assembly | ▪ Using the manufacturer approved diagnostic system, check the snapshot data for details of the missing message. Check the body control module/gateway module assembly for related DTCs and refer to the relevant DTC index |
| U0146-87 | Lost Communication With Gateway "A" - | ▪ Missing message from the body | ▪ Using the manufacturer approved diagnostic |

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| | Missing message | control module/gateway module assembly | system, check the snapshot data for details of the missing message. Check the body control module/gateway module assembly for related DTCs and refer to the relevant DTC index |
| U0147-87 | Lost Communication With Gateway "B" - Missing message | ▪ Missing message from the body control module/gateway module assembly | ▪ Using the manufacturer approved diagnostic system, check the snapshot data for details of the missing message. Check the body control module/gateway module assembly for related DTCs and refer to the relevant DTC index |
| U0291-87 | Lost Communication With Gear Shift Control Module B - Missing message | ▪ Missing message from the transmission control switch | ▪ Using the manufacturer approved diagnostic system, check the snapshot data for details of the missing message. Check the transmission control switch for related DTCs and refer to the relevant DTC index |
| U0300-00 | Internal Control Module Software Incompatibility - No sub type information | ▪ The anti-lock brake system control module is not compatible with the vehicle | ▪ Check the anti-lock brake system control module part number, install the correct part as required. Clear DTC and retest. To confirm the repair and extinguish the warning lamps, clear the DTC, cycle the ignition state to off, then return the ignition state to on and wait up to 10 seconds |
| U0401-81 | Invalid Data Received from ECM/PCM A - Invalid serial data received | ▪ Invalid data received from the engine control module | ▪ Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index |

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| U0401-82 | Invalid Data Received from ECM/PCM A - Alive / sequence counter incorrect / not updated | ▪ Invalid data received from the engine control module | ▪ Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index |
| U0401-83 | Invalid Data Received from ECM/PCM A - Value of signal protection calculation incorrect | ▪ Invalid data received from the engine control module | ▪ Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index |
| U0402-81 | Invalid Data Received from TCM - Invalid serial data received | ▪ Invalid data received from the transmission control module | ▪ Using the manufacturer approved diagnostic system, check the transmission control module for related DTCs and refer to the relevant DTC index |
| U0402-82 | Invalid Data Received from TCM - Alive / sequence counter incorrect / not updated | ▪ Invalid data received from the transmission control module | ▪ Using the manufacturer approved diagnostic system, check the transmission control module for related DTCs and refer to the relevant DTC index |
| U0402-83 | Invalid Data Received from TCM - Value of signal protection calculation incorrect | ▪ Invalid data received from the transmission control module | ▪ Invalid data received from the transmission control module |
| U0403-81 | Invalid Data Received From Transfer Case Control Module - Invalid serial data received | ▪ Invalid data received from the transfer case control module | ▪ Using the manufacturer approved diagnostic system, check the transfer case control module for related DTCs and refer to the relevant DTC index |
| U0404-81 | Invalid Data Received from Gear Shift Control Module A - Invalid serial data | ▪ Invalid data received from the transmission control switch | ▪ Using the manufacturer approved diagnostic system, check the transmission control |

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| | received | | switch for related DTCs and refer to the relevant DTC index |
| U0404-82 | Invalid Data Received from Gear Shift Control Module A - Alive / sequence counter incorrect / not updated | ▪ Invalid data received from the transmission control switch | ▪ Using the manufacturer approved diagnostic system, check the transmission control switch for related DTCs and refer to the relevant DTC index |
| U0404-83 | Invalid Data Received from Gear Shift Control Module A - Value of signal protection calculation incorrect | ▪ Invalid data received from the transmission control switch | ▪ Using the manufacturer approved diagnostic system, check the transmission control switch for related DTCs and refer to the relevant DTC index |
| U0405-68 | Invalid Data Received From Cruise Control Module - Event information | ▪ Missing/invalid data from the adaptive speed control module | ▪ Using the manufacturer approved diagnostic system, check the adaptive speed control module for related DTCs and refer to the relevant DTC index |
| U0414-81 | Invalid Data Received From Four-Wheel Drive Clutch Control Module - Invalid serial data received | ▪ Invalid data received from the transfer case control module | ▪ Using the manufacturer approved diagnostic system, check the transfer case control module for related DTCs and refer to the relevant DTC index |
| U0414-82 | Invalid Data Received From Four-Wheel Drive Clutch Control Module - Alive / sequence counter incorrect / not updated | ▪ Invalid data received from the transfer case control module | ▪ Using the manufacturer approved diagnostic system, check the transfer case control module for related DTCs and refer to the relevant DTC index |
| U0414-83 | Invalid Data Received From Four-Wheel Drive Clutch Control Module - Value of | ▪ Invalid data received from the transfer case control module | ▪ Using the manufacturer approved diagnostic system, check the transfer case control |

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| | signal protection calculation incorrect | | module for related DTCs and refer to the relevant DTC index |
| U0428-81 | Invalid Data Received From Steering Angle Sensor Module - Invalid serial data received | ▪ Invalid data received from the steering angle sensor module | ▪ Using the manufacturer approved diagnostic system, check the steering angle sensor module for related DTCs and refer to the relevant DTC index |
| U0428-82 | Invalid Data Received From Steering Angle Sensor Module - Alive / sequence counter incorrect / not updated | ▪ Invalid data received from the steering angle sensor module | ▪ Using the manufacturer approved diagnostic system, check the steering angle sensor module for related DTCs and refer to the relevant DTC index |
| U0428-83 | Invalid Data Received From Steering Angle Sensor Module - Value of signal protection calculation incorrect | ▪ Invalid data received from the steering angle sensor module | ▪ Using the manufacturer approved diagnostic system, check the steering angle sensor module for related DTCs and refer to the relevant DTC index |
| U042B-81 | Invalid Data Received from Chassis Control Module "A" - Invalid serial data received | ▪ Invalid data received from the integrated suspension control module | ▪ Using the manufacturer approved diagnostic system, check the integrated suspension control module for related DTCs and refer to the relevant DTC index |
| U0432-82 | Invalid Data Received From Multi-axis Acceleration Sensor Module - Alive / sequence counter incorrect / not updated | ▪ Invalid data received from the restraints control module | ▪ Using the manufacturer approved diagnostic system, check the restraints control module for related DTCs and refer to the relevant DTC index |
| U0432-83 | Invalid Data Received From Multi-axis Acceleration Sensor Module - Value of | ▪ Invalid data received from the restraints control module | ▪ Using the manufacturer approved diagnostic system, check the restraints control module |

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| | signal protection calculation incorrect | | for related DTCs and refer to the relevant DTC index |
| U0437-81 | Invalid Data Received From Differential Control Module - Rear - Invalid serial data received | ▪ Invalid data received from the rear differential control module | ▪ Using the manufacturer approved diagnostic system, check the rear differential control module for related DTCs and refer to the relevant DTC index |
| U0442-81 | Invalid Data Received From ECM/PCM "B" - Invalid serial data received | ▪ Invalid data received from the engine control module | ▪ Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index |
| U0447-00 | Invalid Data Received From Gateway "A" - No sub type information | ▪ Invalid data received from the body control module/gateway module assembly | ▪ Using the manufacturer approved diagnostic system, check the body control module/gateway module assembly for related DTCs and refer to the relevant DTC index |
| U0447-54 | Central Configuration - Missing calibration | ▪ Car configuration file mismatch with vehicle specification | ▪ Using the manufacturer approved diagnostic system, re-configure the anti-lock brake system control module with the latest level software. If the fault persists, check and up-date the car configuration file as necessary |
| U0513-82 | Invalid Data Received From Yaw Rate Sensor Module - Alive / sequence counter incorrect / not updated | ▪ Invalid data received from the restraints control module | ▪ Using the manufacturer approved diagnostic system, check the restraints control module for related DTCs and refer to the relevant DTC index |
| U0513- | Invalid Data Received | ▪ Invalid data received | ▪ Using the manufacturer |

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| 83 | From Yaw Rate Sensor Module - Value of signal protection calculation incorrect | from the restraints control module | approved diagnostic system, check the restraints control module for related DTCs and refer to the relevant DTC index |
| U0536-81 | Invalid Data Received From Lateral Acceleration Sensor Module - Invalid serial data received | ▪ Invalid data received from the restraints control module | ▪ Using the manufacturer approved diagnostic system, check the restraints control module for related DTCs and refer to the relevant DTC index |
| U0592-81 | Invalid Data Received From Gear Shift Control Module B - Invalid serial data received | ▪ Invalid data received from the transmission control switch | ▪ Using the manufacturer approved diagnostic system, check the transmission control switch for related DTCs and refer to the relevant DTC index |
| U0595-81 | Invalid Data Received From Powertrain Control Monitor Module - Invalid serial data received | ▪ Invalid data received from the engine control module | ▪ Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index |
| U101C-87 | Lost Communication with PCM Node D - Missing message | ▪ Engine control module power or ground circuit open circuit, high resistance<br>▪ High speed CAN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Engine control module system fault | ▪ Refer to the electrical circuit diagrams and check the engine control module power and ground circuits for open circuit, high resistance<br>▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Using the manufacturer approved diagnostic |

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| | | | system, check the engine control module for related DTCs and refer to the relevant DTC index |
| U101D-87 | Lost Communication with PCM Node E - Missing message | ▪ Engine control module power or ground circuit open circuit, high resistance<br>▪ High speed CAN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Engine control module system fault | ▪ Refer to the electrical circuit diagrams and check the engine control module power and ground circuits for open circuit, high resistance<br>▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index |
| U101E-87 | Lost Communication with PCM Node J - Missing message | ▪ Engine control module power or ground circuit open circuit, high resistance<br>▪ High speed CAN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Engine control module system fault | ▪ Refer to the electrical circuit diagrams and check the engine control module power and ground circuits for open circuit, high resistance<br>▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance<br>▪ Using the manufacturer approved diagnostic system, check the engine control module for |

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| | | | related DTCs and refer to the relevant DTC index |
| U2101-55 | Control Module - Not configured | <ul><li>The anti-lock brake system control module is not compatible with the vehicle</li><li>Anti-lock brake system control module is not correctly configured</li></ul> | <ul><li>Check the anti-lock brake system control module part number, install the correct part as required</li><li>Using the manufacturer approved diagnostic system, re-configure the anti-lock brake system control module with the latest level software. Clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module</li></ul> |
| U3000-44 | Control Module - Data memory failure | <ul><li>Anti-lock brake system control module failure</li></ul> | <ul><li>Using the manufacturer approved diagnostic system, re-configure the anti-lock brake system control module with the latest level software. If the fault persists, install a new anti-lock brake system control module</li></ul> |
| U3000-46 | Control Module - Calibration/parameter memory failure | <ul><li>Anti-lock brake system control module internal failure</li></ul> | <ul><li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module</li></ul> |
| U3000-4B | Control Module - Over temperature | <ul><li>Anti-lock brake system hydraulic control unit valve overheat protection has been activated</li></ul> | **NOTE:**<br>This DTC may be set if the manufacturer approved diagnostic system has been operating the valves for a prolonged period |

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| | | | ▪ Allow sufficient time for the unit to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest |
| U3000-51 | Control module - Not programmed | ▪ Anti-lock brake system control module is not correctly configured | ▪ Using the manufacturer approved diagnostic system, re-configure the anti-lock brake system control module with the latest level software. Clear the DTCs and retest. If the fault persists, install a new anti-lock brake system control module |